

**UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION**

Interconnection of Large Loads to the	)	Docket No. RM26-4-000
Interstate Transmission System	)	

Reply Comments of GridCARE

GridCARE respectfully submits these reply comments in response to initial comments filed on the Commission’s Advanced Notice of Proposed Rulemaking (ANOPR) on Large Load and Hybrid Interconnections. We appreciate the Commission’s leadership in recognizing that the current interconnection framework is not well-suited for the rapidly evolving landscape of large, hybrid, and flexible loads, and examining a framework to ensure timely, orderly, and beneficial interconnection of large loads.

GridCARE is a U.S. based company that provides utilities, system operators, and large-load customers with a secure platform capable of identifying and unlocking latent capacity on the grid - capacity that is already built but underutilized. Through AI-enabled temporal analysis, probabilistic modeling, and verifiable load flexibility, GridCARE enables faster interconnection pathways for large loads and drives beneficial load growth, thereby creating a bridge between the time needed to build new infrastructure and resources available to pay for it, while also aligning with national AI, economic, and security goals.

Across the record, numerous commenters highlight themes that GridCARE strongly supports, including:

- Better grid utilization can enable faster interconnections without compromising reliability or long-term planning needs;
- Modernized planning is essential to understanding latent grid capacity and the role of flexibility; and
- An expedited or “fast-track” study path for projects able to demonstrate verifiable flexibility is both feasible and beneficial.

These themes are consistent with regional efforts already underway and reflect a constructive path for the Commission’s framework.

II. Background

Several commenters noted that longstanding planning and interconnection methodologies—relying heavily on static load profiles, worst-case peak assumptions, and deterministic single-snapshot analyses—no longer reflect how the modern grid operates nor how large loads actually behave. These methods were designed for an era dominated by centralized generation and predictable demand patterns. They do not capture the temporal

diversity, non-coincident peaks, thermal behavior of assets, or the flexible operational characteristics of today's large loads, including AI data centers and other digital infrastructure facilities.

As a result, planners can underestimate the hosting capacity that exists across the vast majority of hours. This gap between actual operational conditions and modeling assumptions contributes to the perception that only long-lead infrastructure can serve new load, when in many cases the constraint may be limited to a small number of hours or conditions.

Comments also reflect that utilities and RTOs lack standardized or consistent formats for customer-provided data, making it challenging to assess large numbers of requests or differentiate between short-duration constraints and structural limitations. These process challenges—not physical scarcity—frequently limit access to existing capacity.

A practical example of improved approaches is emerging through GridCARE's collaboration with Portland General Electric (PGE). Using advanced analytics, PGE and GridCARE identified and validated meaningful amounts of latent capacity on existing transmission and distribution infrastructure. As publicly reported by PGE,¹ more than 80 MW of additional capacity is expected to be available for new large loads in 2026, increasing over time as additional operational insights are incorporated. In a region where conservative study assumptions historically extended interconnection timelines, this work has helped PGE refine planning models and identify near-term capacity with greater confidence.

This analytical foundation has supported early projects, such as the recently announced flexible energy deployment by Aligned Data Centers and Calibrant Energy in Hillsboro, Oregon²—an example of how utilities and developers can coordinate flexibility and operational data to accelerate new load connections while maintaining system reliability and affordability.

To illustrate the broader potential of latent capacity, GridCARE conducted representative modeling demonstrating that interconnecting large AI loads through existing infrastructure—where operationally feasible—can reduce systemwide costs by improving asset utilization, increasing utility revenues without imposing new costs on existing customers, and supporting future capital investments. While specific outcomes will vary by region and system topology, the analysis demonstrates that beneficial load growth can occur when interconnection pathways reflect real-world system conditions and flexible operational capabilities.³

By modernizing planning practices, improving visibility into system operations, and enabling a process to bring speed to power, at scale, through incentivizing large load interconnections that are hybrid or willing to be flexible, the United States can unlock capacity that already exists on the system—supporting AI leadership, domestic manufacturing, and national security priorities without compromising reliability and waiting years for new infrastructure to come online. Modernizing interconnection and planning approaches can complement long-term investment rather than replace it, enabling near-term load growth while utilities build the assets needed for the next decade.

¹<https://www.prnewswire.com/news-releases/portland-general-electric-and-gridcare-accelerate-hundreds-of-megawatts-of-data-center-power-in-leading-us-market-302577731.html>

²<https://www.datacenterknowledge.com/energy-power-supply/aligned-to-deploy-first-of-its-kind-data-center-battery-storage-project>

³ www.gridcare.ai/post/ai-data-centers-as-engines-of-affordability-and-capital-investment

III. Initial Comments Support Modernized Interconnection and Latent-Capacity Approaches

The initial comments reflect considerable alignment across diverse stakeholders—RTOs, utilities, data center operators, technology providers, and public interest organizations—on several themes relevant to unlocking latent capacity and incorporating flexibility.

Simultaneous study of load, flexibility, and any co-located resources

Commenters across the spectrum—including PJM, ISO-NE, SPP, Amazon Energy, PG&E, National Grid, and the Energy Storage Coalition—agree that modern large loads cannot be evaluated as static, inflexible facilities. They increasingly come with on-site storage, demand-shaping capabilities, thermal management flexibility, backup or supplemental generation, or the ability to coordinate operations across multiple sites.

Studying load and resources together can reveal significantly different system impacts than studying each independently. Commenters note that this approach better reflects actual operating conditions and can avoid overstating upgrade needs.

Recognizing flexibility in interconnection studies and planning

Many commenters support the Commission's proposal to establish an expedited or alternative pathway for large loads willing and able to operate flexibly. Others highlight that operational flexibility, when verified, may reduce peak-hour impacts and allow planners to distinguish between constraints that are structural versus those that are transient or condition-specific.

GridCARE agrees that flexibility, when defined in measurable and verifiable terms, can help accelerate interconnections, reduce restudy risks, provide near-term operational headroom, and maintain reliability. This does not replace long-term investment; rather, it enables the system to serve new load while that investment is underway.

RTOs and ISOs are already developing frameworks that incorporate flexibility and latent capacity

PJM's Critical Issue Fast Path, SPP's High-Impact Large Load process, and ISO-NE's ongoing work on load characterization all demonstrate that regions are exploring ways to streamline studies, incorporate temporal and behavioral data, better align study processes with actual system operations, and improve coordination with utilities.

GridCARE supports the Commission enabling this direction through a national framework while allowing regional entities to tailor implementation.

Higher grid utilization supports affordability and protects current customers

Public interest organizations, state commissions, and utilities highlight the need to prevent unnecessary cost shifts. Better identification of latent capacity and more precise characterization of load behavior can help ensure that upgrades are only required where they are truly necessary.

IV. Recommendations

Based on the record and GridCARE's experience working with utilities and large-load customers, GridCARE respectfully offers the following recommendations:

1. Establish a national framework that recognizes flexibility and modern operational characteristics.

The comprehensive portfolio of resources and technologies now available—ranging from front-of-the-meter and co-location flexibility (like utility-scale storage and fast-ramping generation) and energy storage (grid-side and customer-side battery systems) to coordinated behind-the-meter generation (backup generators and CHP) and advanced transmission technologies (Dynamic Line Ratings, PSTs)—collectively form a significant and valuable flexibility ecosystem that can be orchestrated to reduce net loading on critical elements during peak or emergency conditions, shape and cap large load demand, and unlock additional transfer capability. Systematically incorporating this flexibility ecosystem into planning and operations will unlock existing capacity on the grid, accelerate interconnections, and enable beneficial load growth to preserve customer affordability and fund new infrastructure investment.

Realizing the value of this flexibility ecosystem requires a transparent, consistent national framework, with high-level expectations for incorporating temporal, behavioral, and operational data into studies, rules and tariffs. The Commission can leverage its approach in establishing the framework and principles of measurement, telemetry, verification, and performance-based obligations in Order No. 2222 for non-traditional distributed resources to guide a modernized, technology-neutral approach for flexible large-load interconnections, while retaining regional discretion to customize methodologies.

2. Encourage a First-Served, Flex-Ready Cluster Study Process

Several transmission providers have begun implementing cluster studies for large load interconnections. However, these analyses do not offer a proactive, system-wide view of transmission headroom in high demand areas. Cluster studies themselves do not create additional capacity or materially accelerate large load interconnections; they simply allocate whatever transmission capacity already exists.

To improve transparency and reduce siting uncertainty, the Commission could encourage or require transmission providers to publish transmission level hosting capacity maps or locational headroom indicators that give large load customers signals on where interconnection is most feasible. If proposed loads can demonstrate verifiable operational flexibility, transmission providers should also offer clear estimates of how such flexibility could reduce study duration or accelerate interconnection timelines.

3. Encourage optional expedited pathways for large loads able to demonstrate flexibility

A national framework should recognize that some loads can materially change their system impacts through operational or technological measures. An expedited study and interconnection pathway, subject to verified performance and data requirements, could improve study efficiency, reduce queues and restudies, accelerate beneficial load growth, and help bridge the gap between near-term demand and long-term infrastructure buildout.

While not mandating specific technologies, the Commission can acknowledge that modern planning and operational tools—capable of analyzing hourly and sub-hourly behavior,

non-coincident peaks, and operational interactions and backup generation technologies — can improve the accuracy of interconnection studies, provide overall system benefits, and reduce unnecessary upgrades.

The Commission should also support study approaches that clearly distinguish between non-exporting, load-modifying behind-the-meter resources and true generation resources seeking firm transmission rights. Applying generator-level deliverability assumptions to non-export configurations can overstate impacts, increase upgrade requirements, and erode the very flexibility the ANOPR seeks to unlock. This distinction is fully consistent with DOE's proposed principle that hybrid facilities should be studied according to the specific injection and withdrawal rights they request. A non-exporting behind-the-meter storage system, by definition, requests zero injection rights and should therefore be studied as a load-modifying resource rather than as a generator seeking firm deliverability.

At the same time, a modern framework should recognize that these behind-the-meter assets may play a more significant system role in the future. As data center energy efficiency improves and additional headroom becomes available, some non-exporting resources could be capable of providing broader reliability or energy services—subject to safeguards and appropriately scoped future interconnection review. Utilities and system operators already demonstrate the practicality of such approaches; for example, ERCOT permits limited-export and non-firm injection configurations for hybrid and storage resources, providing a workable template for how controlled export rights can be managed without triggering full deliverability requirements.⁴ Designing study methods that differentiate these configurations today, while anticipating pathways for optional future system contributions will reduce uncertainty, avoid unnecessary costs, and preserve incentives for large loads to invest in flexibility that supports both interconnection efficiency and system reliability.

4. Establish load-based study fees and performance incentives to improve queue discipline and accelerate interconnection timelines

Consistent with DOE Principle #4 and the study-fee structure adopted in Order No. 2023 for generator interconnections, the Commission should establish a load-interconnection study-fee framework that scales with the size and complexity of the proposed load and reflects the level of operational flexibility demonstrated by the customer. Such fees would discourage duplicative or speculative requests, reduce inflated load forecasts, and ensure that study resources are aligned with actual system impacts.

In parallel, the Commission should consider performance-based incentives for transmission providers tied to each megawatt of load interconnected on an accelerated or performance-defined timeline. Providing timely, flexible large-load interconnections delivers systemwide benefits, including improved grid utilization, increased utility revenues without shifting costs to existing customers, and faster realization of national economic and security priorities. Incentives tied to verified outcomes would encourage utilities to adopt modernized

⁴ * For example, ERCOT allows hybrid and storage resources to interconnect under export-limited or non-firm injection configurations through controllable inverter settings and plant-level telemetered limits. See, e.g., ERCOT Nodal Protocols §§ 3.3.1 (Resource Interconnection), 3.15 (Generic Transmission Constraints), and ERCOT Planning Guide § 5.4 (Interconnection of Generation Resources), which outline mechanisms for limited-export resources, operational caps on injection, and curtailment of non-firm exports under system conditions. These constructs demonstrate that controlled export rights can be safely accommodated without requiring full deliverability treatment.

study approaches and leverage flexibility pathways where appropriate without impacting system reliability.

5. Reinforce that latent capacity and flexibility complement, but do not replace, transmission and resource development

Numerous commenters emphasize—and GridCARE agrees—that long-term resource adequacy and transmission expansion remain essential. A framework that better utilizes existing assets can serve near-term load, maintain affordability, and support the investment environment for utilities while long-lead projects progress through permitting and construction.

6. Allow RTOs/ISOs and utilities to tailor implementation

Given regional diversity in grid structure, generation mix, and regulatory frameworks, implementation should be grounded in regional stakeholder processes, subject to Commission oversight. FERC's role should be to establish a consistent structure and principles, while enabling flexibility in how regions meet them.

V. Conclusion

The record demonstrates strong support for modernizing interconnection frameworks to reflect the operational realities of today's large loads and the evolving grid. Approaches that incorporate temporal behavior, flexible operations, and co-located resources can help unlock latent capacity and accelerate interconnections while maintaining reliability and affordability.

GridCARE appreciates the Commission's leadership in advancing this discussion and supports a final framework that recognizes flexibility, encourages modern analytical approaches, enables optional expedited pathways, and allows regions to adapt implementation to their specific system needs.

Such an approach can help bridge the gap between immediate large-load requirements and the timelines for building new infrastructure, supporting national economic and security priorities while protecting existing customers and upholding reliability standards.

Respectfully submitted,
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